

3-3 Milestone: Paper Prototype

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A paper prototype of the garden tracking app was developed to enable simulating several actions. It was tested by two potential users who had been initially interviewed previously, and their interactions, thoughts, and impressions were recorded. Results of the user tests are analyzed and modifications to the initial user interface based on their feedback are proposed.

Paper Prototype Development

The original focus of the app was to provide assistance tracking soil nitrogen and moisture levels. During the initial round of user research interview, it became evident that a slightly different focus was more eagerly desired: namely, to track the accomplishment and scheduling of the watering, fertilizing, planting, and harvesting of garden crops. To that end, the paper prototype mock-up was designed to test the layout of a “garden checkup” interaction, in which the user is logging their watering of the garden and recording different notable pieces of information. Beginning with several garden plots and crops preconfigured, this process was sketched out on a series of five screens (see Appendix A).

It began with the home screen, screen 1, consisting of a series of action buttons and a short dashboard of the last time significant garden interactions were accomplished. Screen 2 assumed the user tapped the “Log Watering” action button, and brought up a form for logging which garden plots were watered. Screen 3 depicted tapping a garden plot to specify a specific crop in a plot was not watered. Screen 4 depicted the updated “Log Watering” screen, reflecting the user’s input of a crop not being watered. Screen 5 depicted the user adding a note and several observations about the state of the garden.

Two additional screens were provided, a side-drawer style main menu (screen 6) and a activity log summary page (screen 7). These were included due to one user being particularly

adamant that a universally available main menu button was a must, and because all three interviewees desired an engaging activity log to reflect on their garden's progress. Animation and button explanations were written in blue, and a very simple set of colors for UI elements was depicted.

Interviews

Preparation for the interviews consisted of developing a short introduction statement and practicing how to best utilize the prototype in a live setting. A brief collection of prep notes is enclosed in Appendix B. Two potential users, who had previously been interviewed (one by phone and one in person) were interviewed again for feedback. Each interview was in person and in their homes. A brief introduction was given, explaining the purpose and timeline for the interview, as well as what was specifically being sought after: namely, their thoughts, questions, comments, and concerns while interacting with the prototypes. My final statement was to remember that I was not testing *their* ability to navigate the app, but that *they* were testing the app's ability to be navigated. During the initial round of interviews, I had taken notes on a laptop. During the prototype interviews, however, I used a notepad and pen. The notes from each interview is attached in Appendix C. Each interview lasted around 40 minutes, exceeding the target of 30 minutes due to a preponderance of user suggestions at the end.

The tasks users completed with the prototype and how successful they were

Any challenges your users faced when modeling an interaction with the prototype

Additional observations your users offered when interacting with the prototype

Any changes you would make to additional follow-up interviews

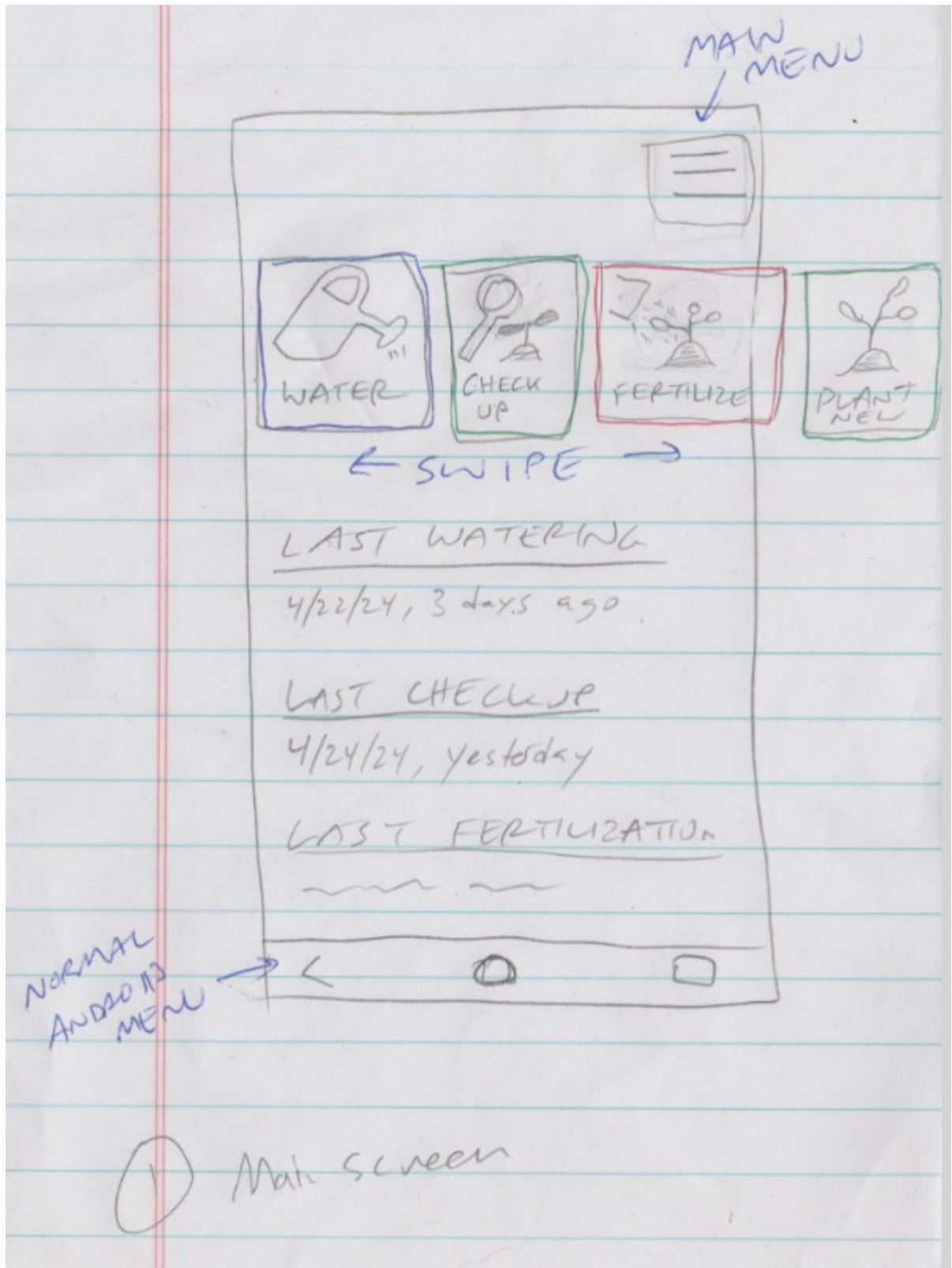
Post-Interview Modifications

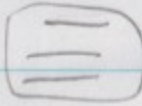
I was somewhat surprised by the difficulty interviewees had with the add notes page. Both commented that seeing a textbox labeled “pests” and another one labeled “fertilizer” underneath a single textbox labeled “Notes” was confusing they did not know why fertilizer was being logged in the “watering” screen, or if noting a fertilizer comment would properly sync to the “Fertilize” option’s data. Feedback regarding button captions and better ordering of main screen activity buttons was also valuable. A critical flaw was also uncovered in which there was no UI element provided to save a note and continue with the process.

In line with the feedback, the following modifications are proposed at this time:

- 1) Restructure the main screen activity buttons to have a large center button in “call-to-action” coloration) to start a Garden Checkup process, and flank this button with smaller buttons for logging a watering, a new planting, and new plot, etc.
- 2) Reword the “Log Watering” screen to something like “New Watering Log Entry”
- 3) Remove the “fertilizer” and “pests” notes from the “Watering” screen

Appendix A: Photocopy of Paper Prototype



LOG WATERING 

Tap to change

WATERED THESE PLOTS

☒ All

☒ East Veggie Garden

☒ Front Twd Flower Garden

☒ Orchard

☒ DONE

< 0 >

add a text note for this watering

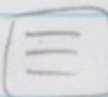
saves the watering log

tap to specify only certain plants were watered

select all plots

2 Logging a watering

* Clicked "all crops" to change

LOG WATERING 

4/25/24 09:26 am

~~WATERED THESE CROPS~~
 WATERED THESE
 CROPS IN East
 Veggie Garden

☐ all
☒ tomatoes
☒ cucumbers
☒ squash
☐ carrots
☒ peas

cancel Save

slides ~~up~~ up

untick the ones that weren't watered

save change and keep logging the watering

③ Tapping all crops to specify, unchecked carrots

LOG WATER

WATERED THESE PLOTS

☐ all

☒ East Veggie Garden

☐ Front Flower Garden

☐ Orchard

This doesn't say "all crops" anymore, but says the crops that were selected

tomatoes, cucumbers, squash, peas

all crops

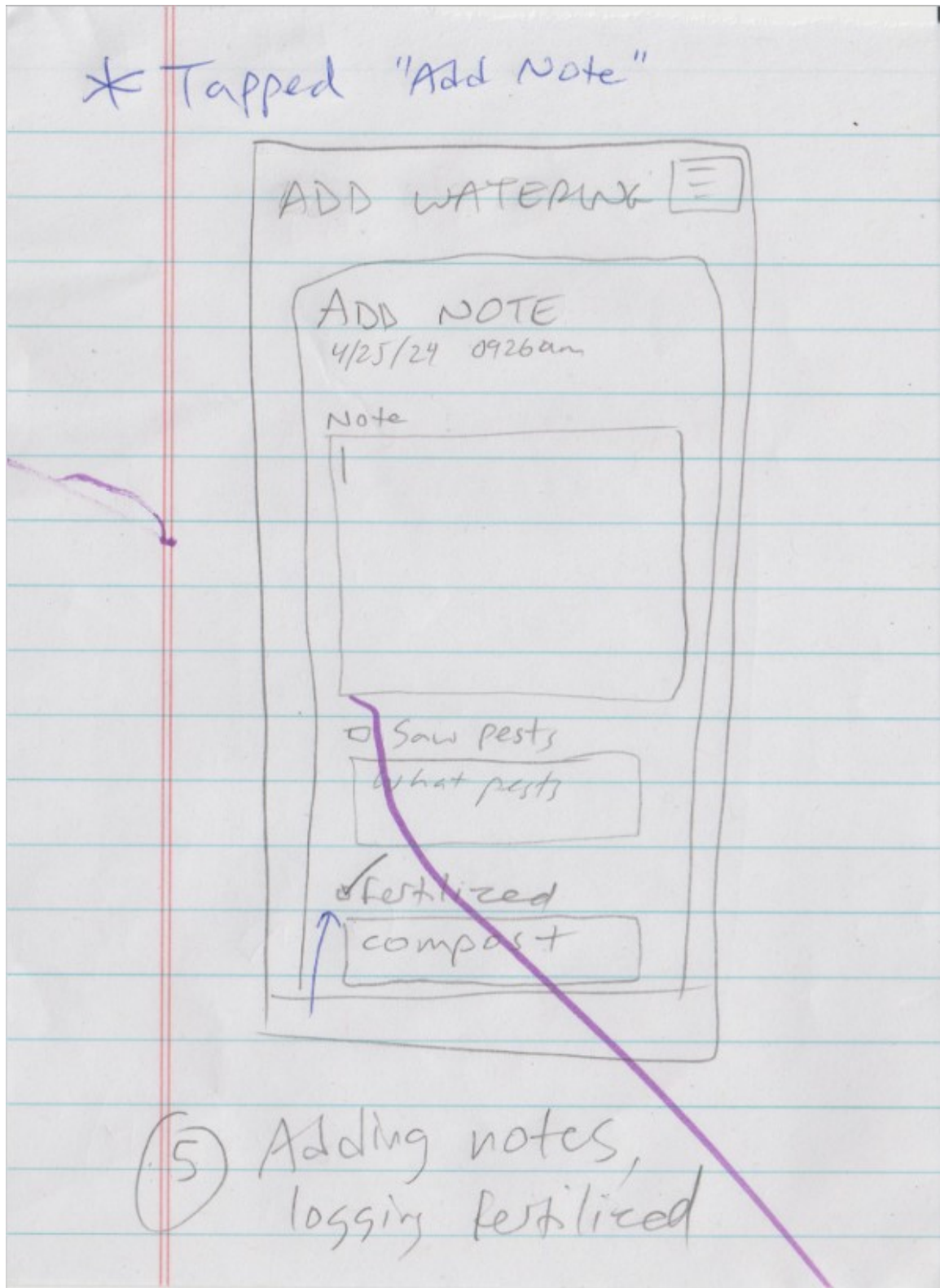
apple tree, peach trees

Add note

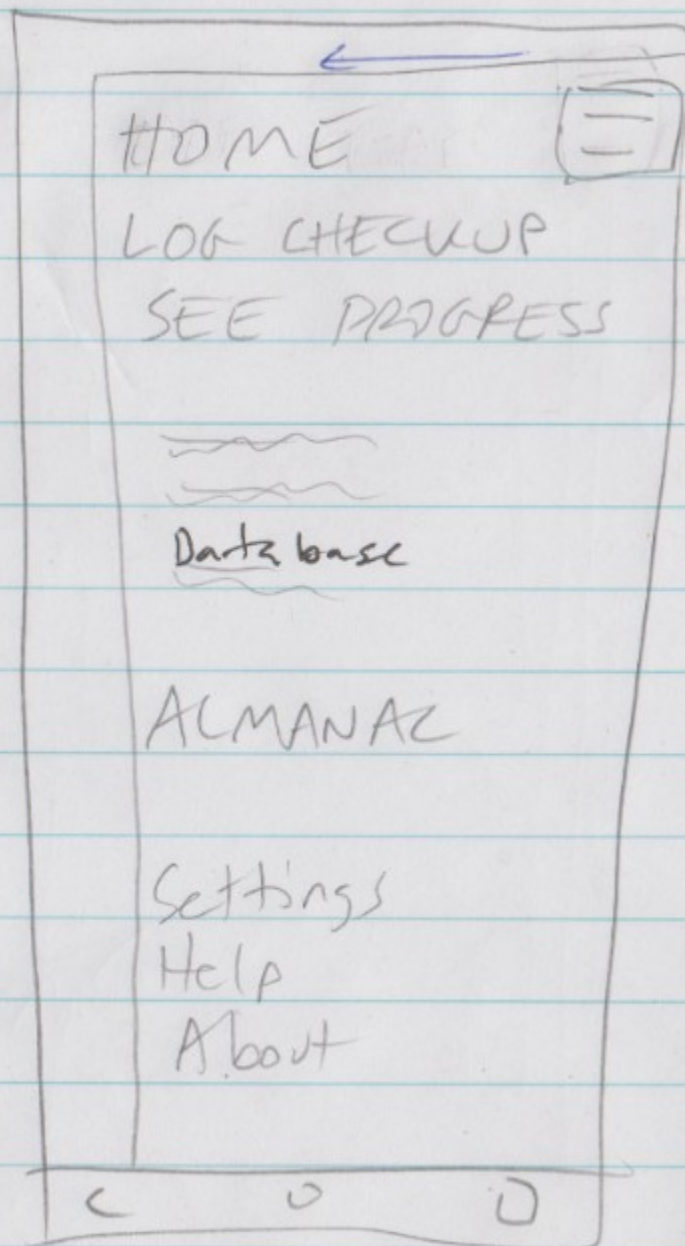
✓ DONE

⏪ ⏩ ⏴ ⏵

④ Logging waterings, after unchecking crops

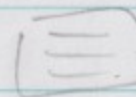


Note: purple line is a stray mark and not intended to represent actual design/layout feature.



(6) MAIN MENU

GARDEN LOG



4/25/24 ☀️

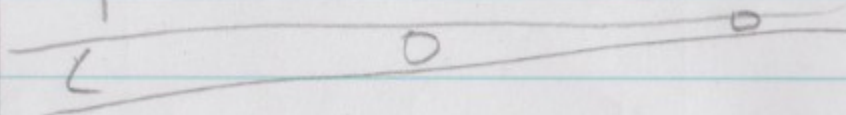
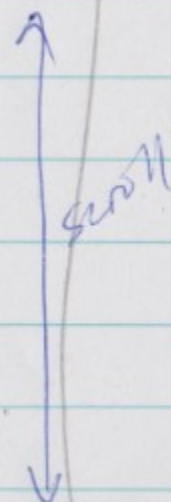
Watered, saw squash beetles
8 tomatoes

4/23/24 ☀️ compost

Watered, added compost
6 tomatoes

4/20/24 ☀️ compost

Watered, composted.
Planted carrots, 1 tomato...



⑦ Log

- ① Main screen
- ② Logging a Watering.
- ③ Selecting crops
- ④ Logging Watering, after changing crop.
- ⑤ Adding notes + fertilizer
- ⑥ Main Menu
- ⑦ Historical Log

Note: Table of contents as an interviewer aid

Appendix B: Interview Preparation

Interview Prep

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Initial instructions

- Good {time}, thank you for doing this follow-up interview with me
- This shouldn't take more than 30 minutes
- The app design is being sketched out and I have paper prototypes to show you and see what you think
- These are just rough sketches of some of the screens
- The idea is that I have you "operate" the app to do a task, giving you different screens as you move through the app
- As you go through the paper screens, please process your thoughts out loud:
 - as you notice things
 - what you are looking around for to continue the process
 - the things you like
 - the things you don't like
 - the things that confuse you, etc.
- The idea is to help me understand what you are thinking, what in the app easy, what isn't, etc

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Interview setting

- At a table, with a notebook for handwritten notes
- paperclipped bunch of "screens" (numbered)
- table of contents to quickly find the right screen

Appendix C: Interview Notes

Interview 1

Interview 1 Notes

- Date: 1/27/2024
- Location: at interviewee's house, while kids were playing in the same room
- Note-taking strategy: notepad, taking notes by hand while interviewee speaks

[Link to handwritten notes](#)

[Edit this section](#)

Transcribed notes

[Edit this section](#)

Introduction/Instructions

- Should last < 30 min
- go over app design paper prototypes
 - you "operate" the app and I give you new screens as you go
- please think out loud:
 - what you notice
 - what you are trying to find
 - what is confusing
 - what you like/dislike
 - any questions you have
 - and suggestions
- finally, please remember: I am not testing you or your abilities.
- instead you and I are both testing the interface

[Edit this section](#)

(1): Main Screen

- [handed out the first paper prototype, explained briefly what the blue text meant]
- ok, if I hit "last watering" then it will bring up more info?
- if I hit water...
- was confused about the main swiping pane with the big buttons

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(2) Adding a Water Log Entry

- wondering what "watering log" (heading) means

[Edit this section](#)

(3) Specifying that one crop was not watered

- ok, unchecked
- -> not water tomato

[Edit this section](#)

(4) Water Screen updated to show the one unwatered crop

- ok, hard to see
- cool
- wondering
- swiping seems good
- if I hit Done
- didn't see the note button

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(5) Adding a Note

- Ok, leaves were wilted
- they recovered from lack of water
- noticed Japanese beetles
- "fertilized", what is that in there
- oh, those are different notes
- the top note is so much bigger
- I like when they give you

- maybe pest, fert, then anything else
- I like direction what to put in there
- more specific note textbox labeled
 - e.g. "note any fungus"
 - then I'm like "oh! I should be keeping track of that"
- it would be so cool to keep track of how many tomatoes, how much fruit I harvested

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(7) Log Overview

- This would be good
- a whole week's worth of
- when did I last fertilize
- I harvested tomato
- I need to check for more tomatoes every day
- I harvested 8 tomatoes today, I can wait a couple days to go back
- or even "plants were wilting on the 23rd, I watered them yesterday, I should check check today, or check for more beetle damage"
- back to home? how do I

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(6) Main Menu

- how to show graphing of water
- plotting # of fruit per day
- I am getting a lot of fruit from this plant
- or this other one
- not down to the individual plant
- from year to year, different
- one year
- connected to the weather app
 - rain

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(1) Main Screen (again)

- I would make them look pretty
- colorful

- Main Menu
 - the icons should be cartoony, colorful, consistent
- Quality Check, Daily Walkthrough, walking the back 40, do some gardening
- "Add new plant" instead of "Plant New"
- the ordering of the menu:
 - Add new plant, Watering, Harvest, then the others
- gardening people tend to b
- fertilizing can be a subheading under watering
 - pots => often fertilize
 - harvest
- Then encyclopedia would have to be pretty generic
- daily notification "have you checked your garden"

Post-interview summary

References to screens are given by screen number in square brackets. For example, [2] refers to the first watering log screen.

- The interviewee expressed some confusion about several UI elements, including:
 - How to operate the main screen activity button scrollable panel [1]
 - The meaning of the heading "Log Watering" [2]
 - How to tell that deselecting a crop for watering was saved [4]
 - Why "pests" and "fertilizer" note textboxes were included in the "watering" workflow, when the home screen had separate fertilizer button [5]
 - How to leave the Log Overview summary [7]
 - The wording of the "Plant New" main screen activity button [1]
- The interviewee successfully navigated the screens, however at several points they wanted to go down a path for which I didn't have a prototype, and so I had to intervene.
- Several observations were proffered by the interviewee:
 - Main screen icons [1] should be cartoony, colorful, and consistent
 - Reword the "Log Watering" screen header [2] to something like "Log a watering" or "Water Log" or "Record watering your garden".
 - Reword the "Check Up" activity button to something like "Daily Walkthrough, Check your garden" or similar [1]
 - Reword the "Plant New" button to something like "Add new plant" [1]
 - Re-order the main screen activity buttons:

- (first to last): Add New Plant, Watering Log, Harvest, and then the rest [1]
 - Re-consider the inclusion of "pests" and "fertilizer" on the Water Log notes screen [5]
 - it is confusing as to why those exist in the watering activity when they have their own buttons on the main screen too
- Changes I would make in subsequent interviews
 - Choose a time/location that is more quiet and secluded to reduce distractions
 - Provide better prompts/explanations on my prototype
 - Be more sparing in my interventions, allowing the user to attempt activities and ponder out loud even if I do not have a prototype for it

Interview 2

Interview 2 Notes

- Date: 1/29/2024
- Location: at interviewee's house, quiet room
- Interview length: 38 minutes
- Note-taking strategy: notepad, taking notes by hand while interviewee speaks, determined to record more notes this time

[PDF handwritten notes](#)

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Transcribed notes

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Introduction/Instructions

- Should last < 30 min
- go over app design paper prototypes
 - you "operate" the app and I give you new screens as you go
- please think out loud:
 - what you notice
 - what you are trying to find
 - what is confusing
 - what you like/dislike
 - any questions you have
 - and suggestions
- finally, please remember: I am not testing you or your abilities.
- instead you and I are both testing the interface

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(1): Main Screen

- what is the squiggle under fertilizer?
- so, when I open fresh w/ no logs yet, do you see "last water: blank"? Probably don't want to do that.
- the icons make sense I think...
- probably should be watering, checkup, fertilized, harvest, plant new
- first time screen should probably say "build your garden"
- actually, more buttons on this screen could be
 - new plot, new garden
 - i should be able to plant a tomato in an existing garden, or in a new garden
 - I plant things in March, but only 1 garden
 - later on I'll add more gardens

- I'm clicking Water

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(2) Adding a Water Log Entry

- "...log...watering." I would swap them around, "watering log"
- so I might do... hmm... so for the date/time, I'd be a, hit the button for date, and a calendar it's already selected for today, and a button at the bottom for accept.
 - and time would be similar, but like a dial or a scroll
- time is necessary as an input because watering @ different times might mean slugs
- ok, so I could select that I watered everything
 - or that I watered each garden
- for the button underneath each plot,
 - rather than saying "all crops"... it could say, like, "specify crops", or, hmm... the wording is a little confusing on that.
 - I'm not sure how to change that
- and then adding a note for the gardening...
- I guess "all crops" makes sense, but the tutorial might have to teach you to "tap the 'all caps' to change what was watered"
- It would be like a drop down menu

[Edit this section](#)

(3) Specifying that one crop was not watered

- [I explained the animation]
- so I could do all, or unselect the ones that weren't watered
- ok, makes sense

[Edit this section](#)

(4) Water Screen updated to show the one unwatered crop

- ok yep, that makes sense
- cool, that looks good
- I still think you need the tutorial the first tome
- I want to click Done
- [intervention] slick add note

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(5) Adding a Note

- clasped hands
- the one thing I would say about this is...is...if you're going to say add fertilizer herem then you need to save it under the fertilized tab too
- it's confusing
- it's almost not the place for it
- the note should just be about watering
- the fertilized and pests shouldn't be here
- "did drip irrigation for 2 hours", "ran the sprinkler 2 hours", "noticed it was very dry" or "the plant's didn't need much water"
- the pests + fert should be in the checkup or fert tab
- I'm worried if I put a fertilized note here, that it won't show up under the fertilized tab [on the main screen]
- I mean, I guess you could add fertilizer in here, because there is certainly time I do both water and fertilized
- if I log it here it must be logged in fertilizer too
- maybe when you enter it here a note pops up saying "do you want it copied to the fert tab too?"
- where is the save button for the add note?
- so if I'm adding a note in watering I only want to add a watering note, not a pests or compost
- fertilizer kind of makes sense because some fertilizer is added to the water
- there should not be a pests or compost on here
- if you check the fertilizer, a drop-down should appear for what kind of fertilizer did you use
- then I click save, the "add note" button should now say "edit note" so you know your note was saved
- and then on the, you click done, I expect you go back to the mainmenu, and the date, are updated for watering, fertilizer, and I expect some kind of icon saying there is a note, or something like "see note" in a different color, and you tap it and the note slides up from the bottom

- yeah, like "see note"

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(6) Main Menu

- [I need to explain animations]
- so what would the progress be?
- ok, like a table of colored blocks for "harvested", "watered"
- and like a histogram showing "tomato yield" vs "spinach yield, like a ranking of how much you got from each plot
- and you could change time lines, from this day to this day
- Almanac link could take you to outside Farmers' Almanac

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(7) Log Overview

- How do get to it?
- looking at it for awhile
- so some things I'm thinking here
- ok, so like you scroll down and see what you've logged each day
- [30 minutes have elapsed]
- yeah, I like it
- well, what is the amount of data you want to show at a time
- like, show it in weeks, and you hit a down arrow to expand it into days of that week
- but this setup is like another tracking app I have
- maybe, the last 7 days pop up automatically, and below that its weekly summaries
- I want to see what Checkup looks like
- I think it should be similar to watering, so, date + time, and then, I guess the bottom part should be like, oh, like gardening
- oh! oh! it should be
 - observation, and a drop-down menu of all your gardens, and you hit go, and then you can do for that garden

- add note
 - pests seen
 - harvesting
 - watering
- the question is how much overlap
- because you already have fertilizing inside watering
 - that takes away from the purpose of the fertilize button
- or [illegible] that just "let me talk about the state of my crops"
- OR, on the main menu, do you want the first button to actually be Daily Log, and that would be everything
 - watering, fertilizing, pests
- and then next to that, options for if you *only* want to log watering, *only* want to log harvesting
- because otherwise every place is going to have overlap
- confusing
- it's like, "do I add fertilizer to watering?"
- but with Daily Log, for each garden:
 - did you water?
 - any pests?
 - did you fertilize?
 - harvest/plant
 - any notes?
- because yeah there's certainly times I just go set up a sprinkler in the morning, I don't check anything
- but other times I go out to check my garden

Post-interview summary

References to screens are given by screen number in square brackets. For example, [2] refers to the first watering log screen.

- The interviewee expressed some confusion about:
 - The squiggly line under "fertilizer" [1] (sloppy prototype)
 - The "all crops" verbiage when logging a watering [2]
 - Note screen [6] is confusing since it has so many boxes
 - why are fertilizer and pests on a screen for watering?
 - Whether adding a fertilizer log on the watering screen also adds an entry to the Fertilizer screen [6]
 - How to reach the Log Overview screen [7]

- Navigation success
 - User navigated the prototype successfully
 - I did need to direct the user toward the activities that I [[
- Several observations were proffered by the interviewee
 - Recommended having activity buttons for "New Plot" or "New Crop" as they build new gardens throughout the year [1]
 - Reword "Log Watering" to "Watering Log" [2]
 - Have a tutorial to walk users through how to input the watering log
 - No save button when adding note [6]
 - Too much overlap between functions will cause confusion in users
 - Activity log should default to showing daily last 7 days and earlier than that are
- Changes I would make in subsequent interviews
 - better annotations on the prototype explaining animations, where the button will go, etc